

Internet of Things

Lab 1

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Internet of Things (IoT): Smart Umbrella

An umbrella that provides information about the likelihood of rain so that users can make a simple decision about whether to take the umbrella with them as they leave their home



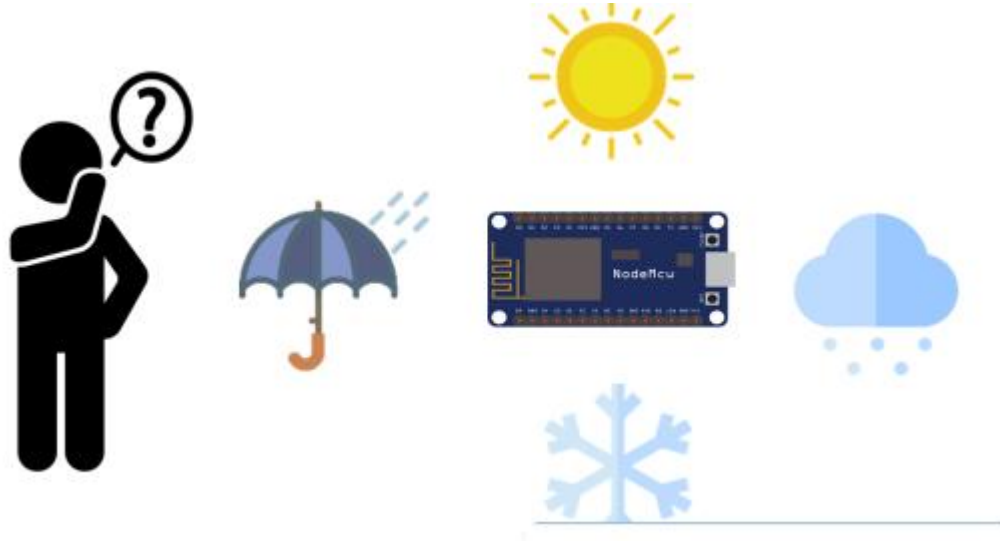
Internet of Things (IoT): Smart Umbrella

The umbrella has a handle that would illuminate when snow or rain was in the forecast.



Internet of Things (IoT): Smart Umbrella

Using existing Wi-Fi technology to pull information about the weather from the Internet



IoT Applications: Quirky Egg Minder

- When your egg supply gets low, this IoT application will send info directly to your phone to remind you to buy more eggs



IoT Applications: WELT

- WELT cares user's overall health by measuring waist size, steps, sitting time and overeating habits with the sensing technology



IoT Applications: Yucky Diaper Sensor

- New startup 24eight has created “wireless diapers” that contain a cellular chip that sends a text message to the lucky mom or dad tasked with cleaning up the mess



Improving Learning Outcomes Using EEG & AI



IoT Applications: HapiFork

- The HapiFork is a Bluetooth-enabled “smart fork” that vibrates when it senses you’re eating too fast.



IoT Applications: Smart Refrigerators

- A refrigerator with a Wi-Fi enabled touch screen that lets you manage your groceries.

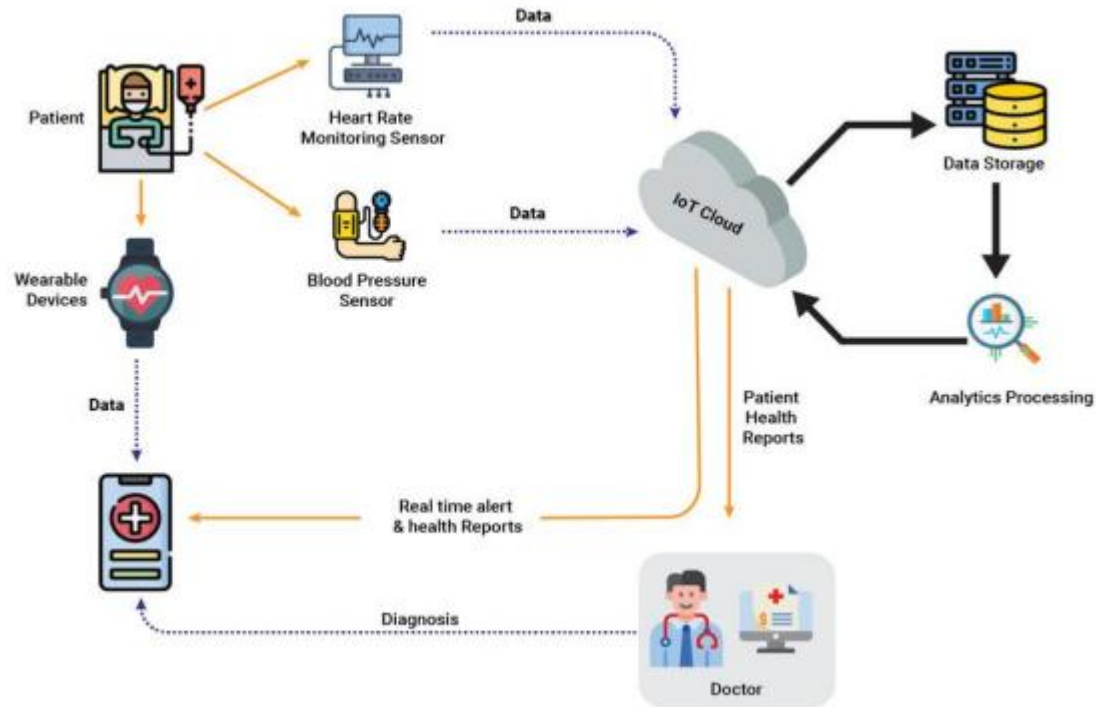


IoT Applications: Smart Shoes

- Smart shoes allow users to change the color of the shoe with one tap on their smartphone.



IoT Applications: Healthcare



Internet Of Things



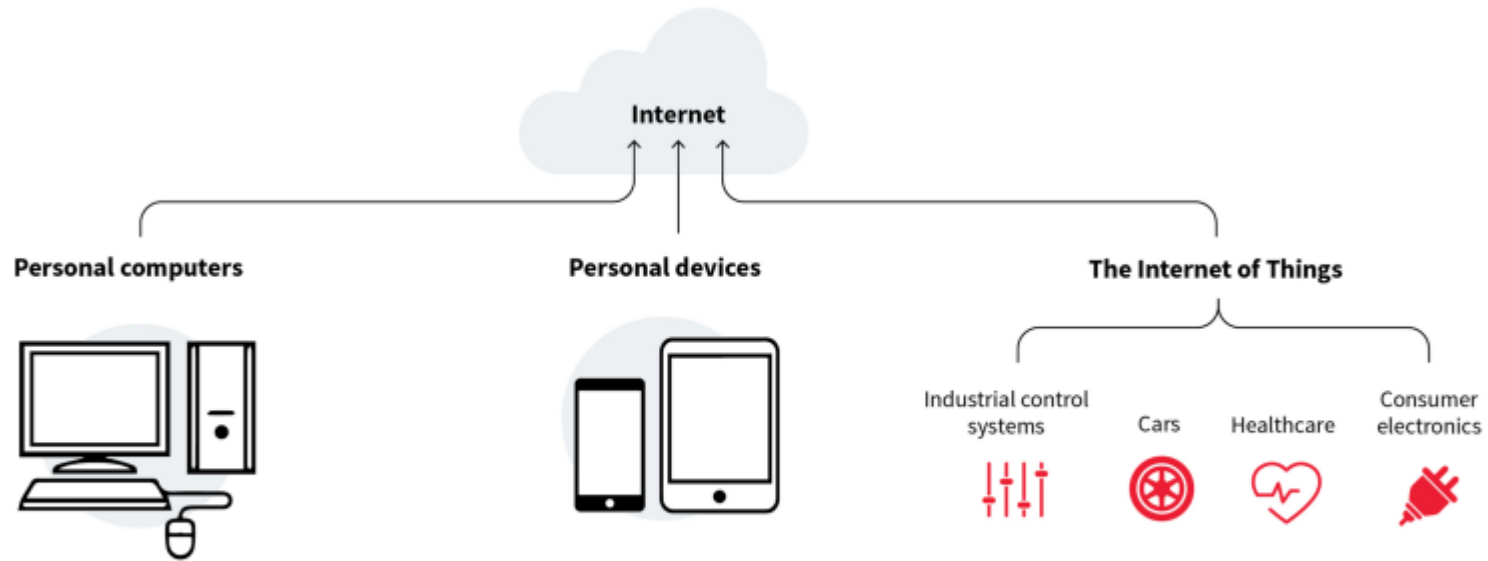
Internet



Network

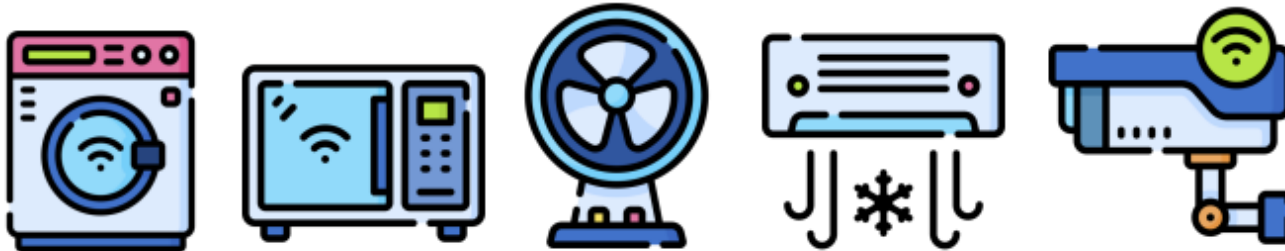


Things



Things

- “Things” are a generic set of entities, including smart devices, sensors, human beings, and any other object that is aware of its context and is able to communicate with other entities, making it accessible at anytime, anywhere.



Things

```
int REDPIN=13;
int YELLOWPIN=12;
int GREENPIN=11;

void setup() {
  //Set the pins to output pins using pinMode
  pinMode(REDPIN, OUTPUT);
  pinMode(YELLOWPIN, OUTPUT);
  pinMode(GREENPIN, OUTPUT);
}

void loop() {
  //Make the LEDs blink in turn:
  //Turn on the red led, and turn off the yellow and green leds.

  digitalWrite(REDPIN, HIGH);
  digitalWrite(YELLOWPIN, LOW);
  digitalWrite(GREENPIN, LOW);
  delay(2000);

  //Turn on the yellow led, and turn off the red and green leds.

  digitalWrite(REDPIN, LOW);
  digitalWrite(YELLOWPIN, HIGH);
  digitalWrite(GREENPIN, LOW);
  delay(2000);

  //Turn on the green led, and turn off the red and yellow leds.

  digitalWrite(REDPIN, LOW);
  digitalWrite(YELLOWPIN, LOW);
  digitalWrite(GREENPIN, HIGH);
  delay(2000);
}
```





Hardware	Quantity
ESP32 Development Board	One
Breadboard	one
Jumper Wires (Male-to-Male (20), Male-to-Female(20), Female-to-Female(10))	20+20+10
USB Cable (Type-A to Micro-USB or USB-C, depending on ESP32 board)	one
LEDs (Assorted Colors, e.g., Red(5), Green(5), Blue(5))	15
Resistors (330Ω(15), 10kΩ(5))	20
Push Buttons or Tactile Switches	2
DHT11 Temperature and Humidity Sensor	1
LDR (Light-Dependent Resistor)	1
Buzzer (Active or Passive)	1
HC-SR04 Ultrasonic Distance Sensor	1

Electronics Stores in Egypt

- LampaTronics
- RAM Electronics
- MicroOhm Electronics
- Makers Electronics
- Alex
- Most Electronics
- Ampere Electronics
- UGE Electronics
- Circuit Electronics
- Electra Store
- ElAbed Electronics
- Free Electronic



AIOT HOME



AIOT HOME

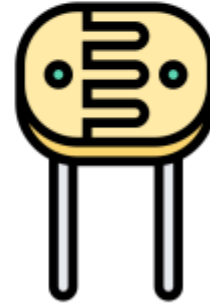


- 01 Main Processor
- 02 Connection Select Switch
- 03 Sensor Block
- 04 LED Block
- 05 GAS Sensor
- 06 GAS Break(Servo Motor)
- 07 Buzzer
- 08 CdS Sensor
- 09 Text LCD
- 10 Camera
- 11 Touch Keypad(3 x 4 key)
- 12 RGB LED
- 13 TFT LCD
- 14 Audio Block
(Sound/Speaker/Mic/Level Bar)
- 15 Temperature/Humidity Sensor
- 16 Door Lock(Servo Motor)
- 17 FAN
- 18 Dust Sensor



Sensors

- A **sensor** is a device that detects some type of input from the physical environment.
- The input can be light, heat, motion, pressure or any number of other environmental phenomena.

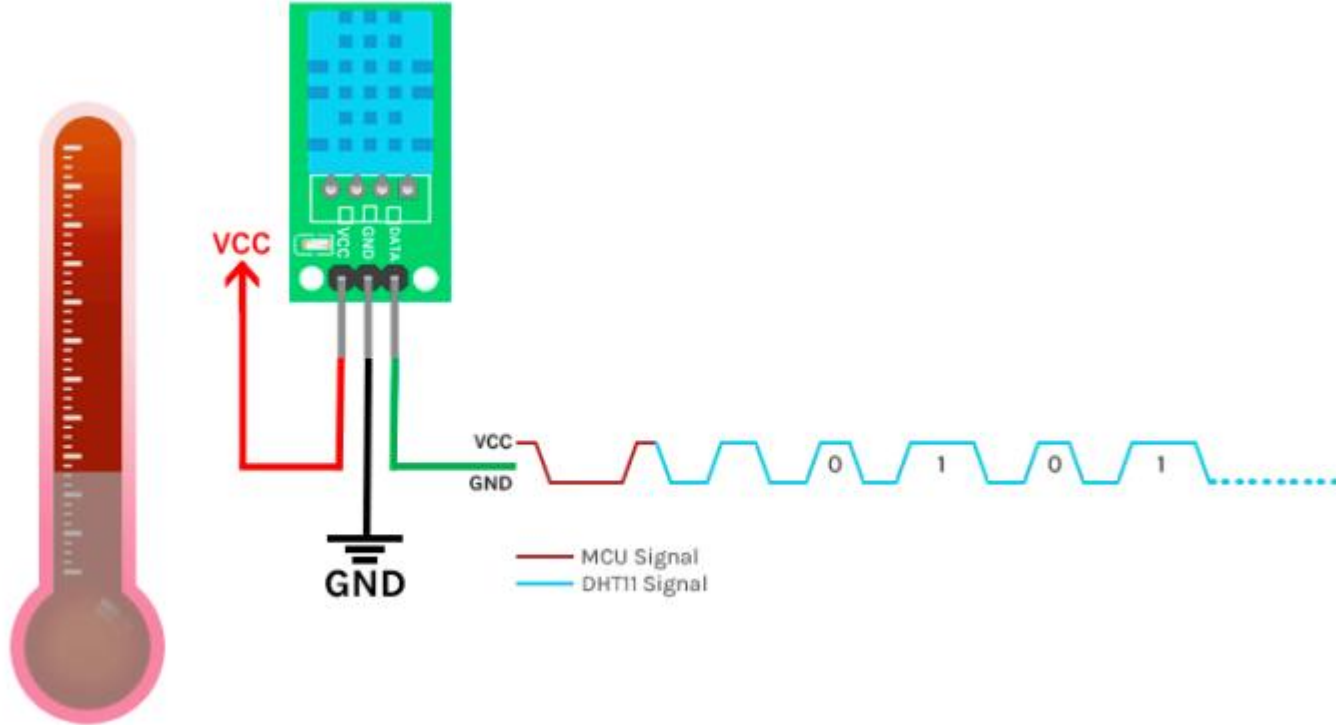


Sensors: Temperature/Humidity Sensor (DHT11)

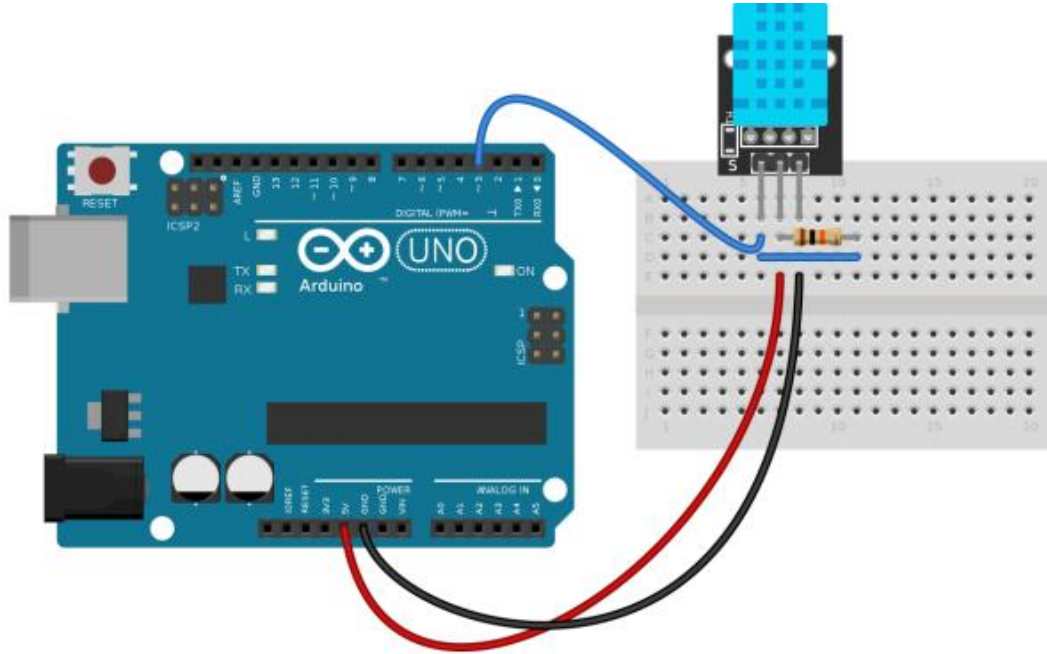
- The **DHT11** sensor measures temperature and humidity.



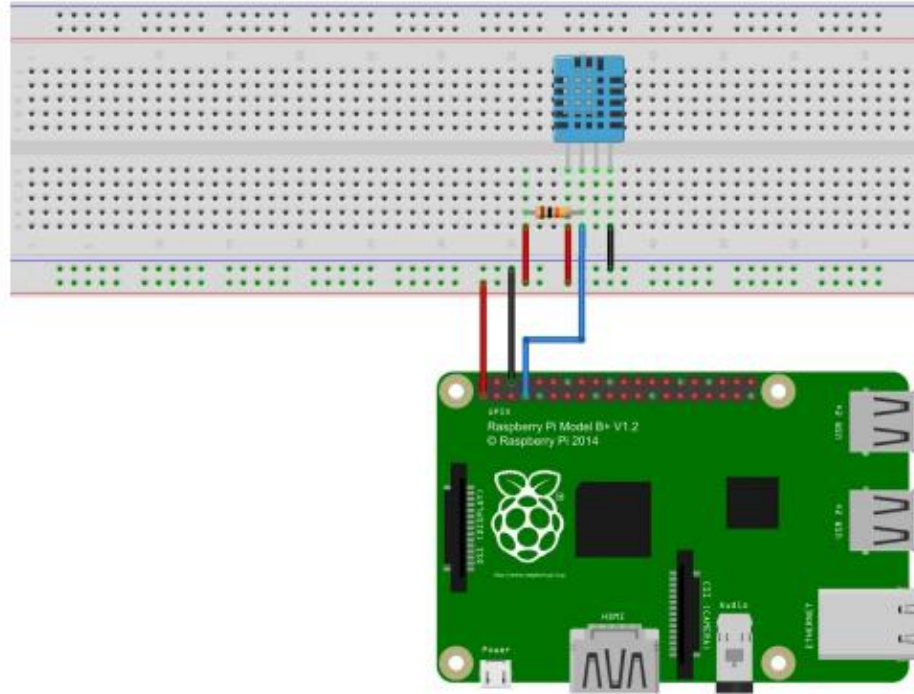
Sensors: Temperature/Humidity Sensor (DHT11)



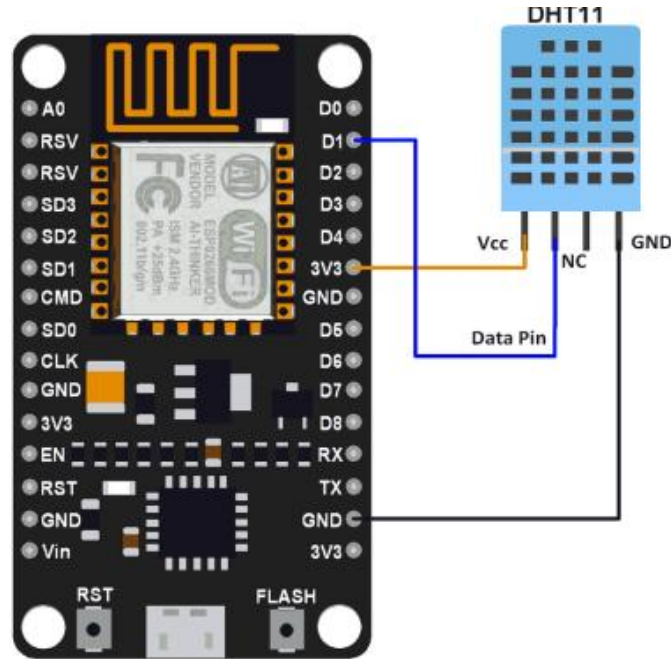
Sensors: Temperature/Humidity Sensor (DHT11)



Sensors: Temperature/Humidity Sensor (DHT11)



Sensors: Temperature/Humidity Sensor (DHT11)



Sensors: PIR Motion Detection Sensor

- The **PIR** (Passive Infrared) sensor allows you to **sense motion**.
- PIR is used to **detect whether a human has moved** in or out of the sensor's range.

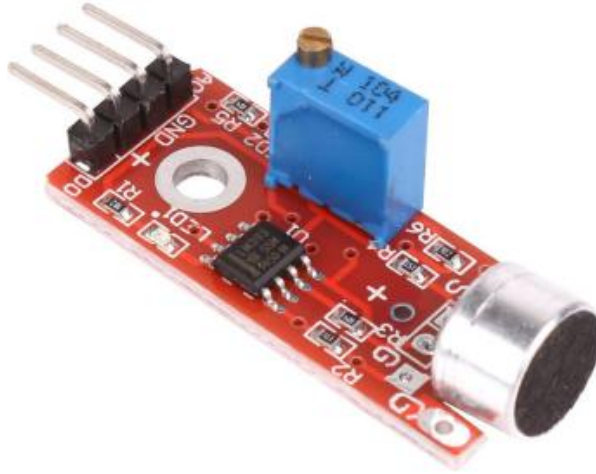


Sensors: PIR Motion Detection Sensor



Sensors: Microphone Sound Detection Sensor

- The **microphone sound sensor**, as the name says, **detects sound**.
- It gives a measurement of **how loud a sound is**.



Sensors: Microphone Sound Detection Sensor



Sensors: Gas Sensor (MQ-2)

- The **MQ-2** gas sensor module is useful for gas leakage detecting.
- The module measures gas such as **butane**.



Sensors: Ultrasonic Sensor (HC-SR04)

- As the name indicates, **ultrasonic sensors** measure **distance** by using **ultrasonic waves**.



Sensors: Ultrasonic Sensor (HC-SR04)



OBJECT



<linuxhint/>



Sensors: Ultrasonic Sensor (HC-SR04)

- The width of the received pulse is used to calculate the distance from the reflected object.
- This can be worked out using the simple distance-speed-time equation we learned in high school.

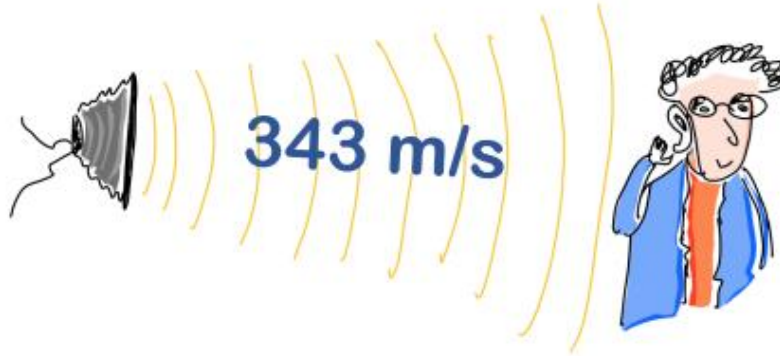


Sensors: Ultrasonic Sensor (HC-SR04)

- For the calculation of the object distance, the sensor measures the time taken by the signal to travel between the transmission of the sound by the transmitter to the reflecting back towards the receiver.

$$\text{Distance} = \frac{1}{2} \text{Time} \times \text{Speed}$$

- The speed of sound in the air at 20°C is 343 m/s.

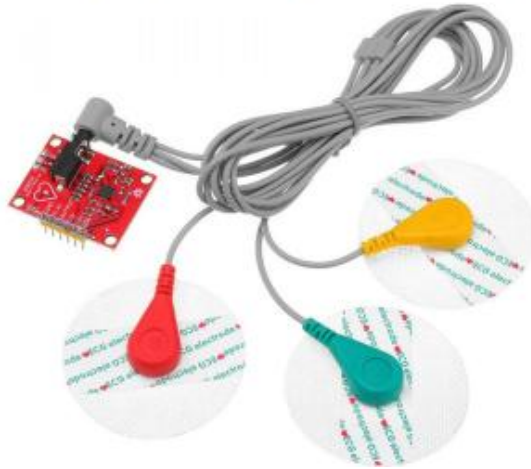


Sensors: Ultrasonic Sensor (HC-SR04)

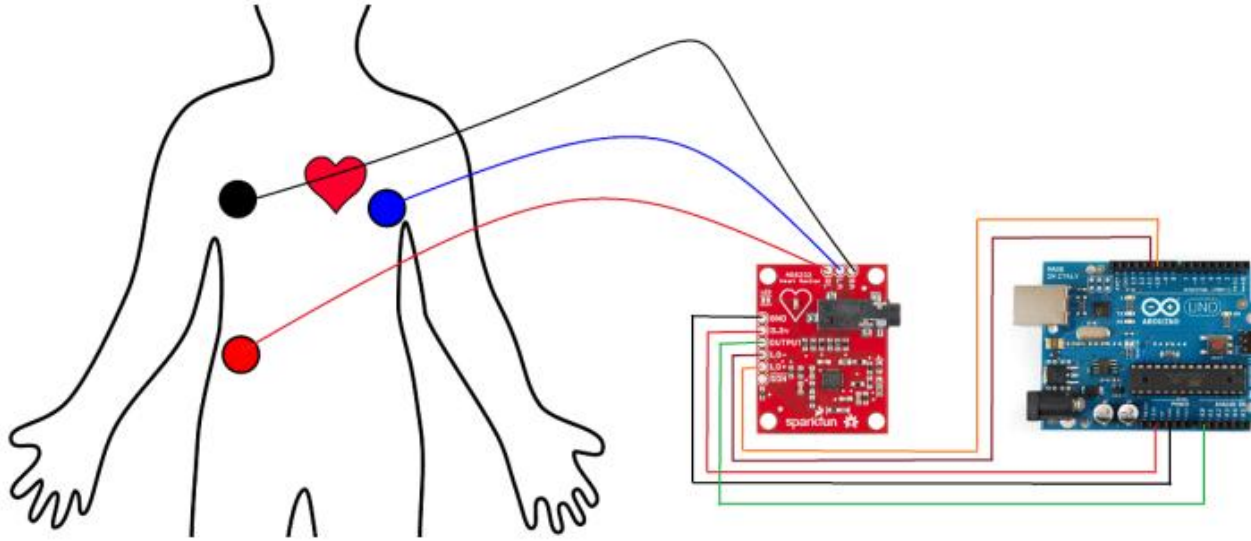


Sensors: ECG AD8232 Heart Rate Sensor

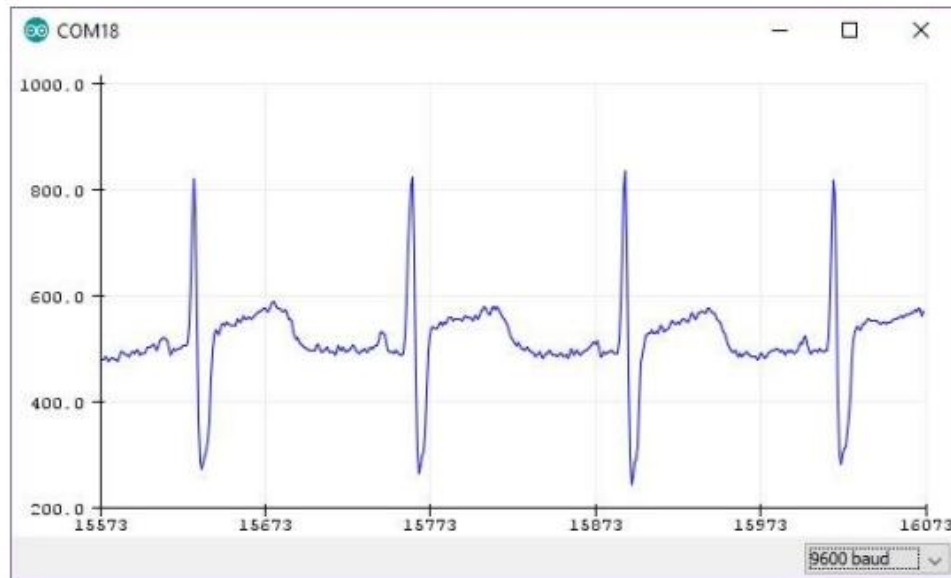
- The heart rate module with the **AD8232** is a device which is capable of **measuring electrical activity of the heart**.
- The activity can be displayed using an **ECG** type graphic.



Sensors: ECG AD8232 Heart Rate Sensor



Sensors: ECG AD8232 Heart Rate Sensor

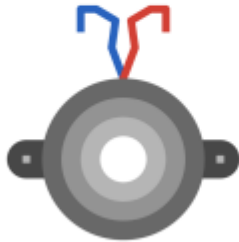


Actuators

- Sensors turn a **physical input** into an electrical output, while **actuators do the opposite**.
- Actuators take electrical signals from control modules and **turn them into physical outputs**.



LEDs



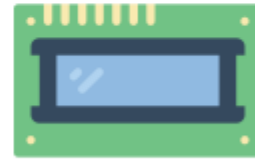
Buzzer



DC Fan



Servo Motor



LCD



Actuators

- A servo motor is an electrical device which can push or rotate an object with great precision.



Actuators

- The HBE-ROBONOVA AI 3 is an intelligent robot with an MR-C3024 controller board capable of controlling 32 servo motors simultaneously.



Getting Started



Arduino Board



Arduino Uno



Arduino Mega



Processing – Raspberry Pi



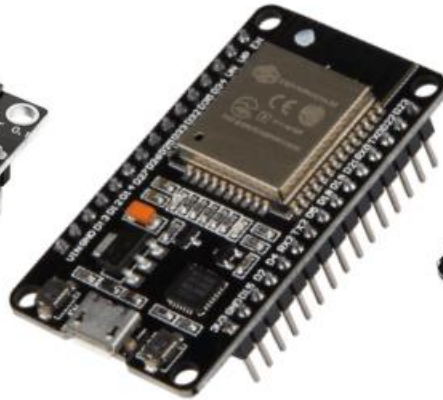
Raspberry Pi



Processing – ESP8266, ESP32 & ESP32-CAM



NodeMCU ESP8266



ESP32



ESP32-CAM

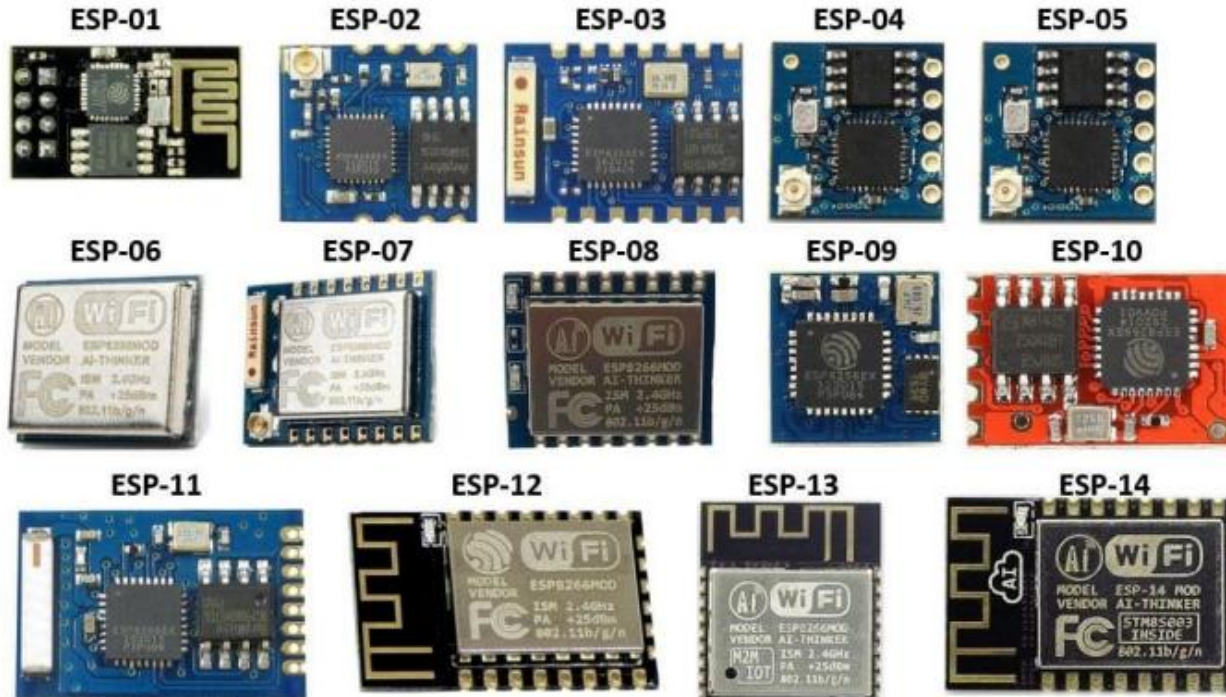


NodeMCU ESP8266 or ESP32?

- The ESP8266 and the ESP32 are **both low-cost** microchips with **Wi-Fi and microcontroller capabilities**, making them well-suited for **IoT applications**.
- **Performance:** The ESP32 has a **more powerful processor** and **more RAM** than the ESP8266.
- **Connectivity:** The ESP32 **supports Bluetooth** in addition to **Wi-Fi**, which makes it better for applications that require **both wireless technologies**.
- **Power consumption:** The ESP32 has a **lower power consumption** than the ESP8266.
- **Price:** The ESP8266 (**270 EGP**) is **cheaper than the ESP32 (430 EGP)**, making it a **more budget-friendly option** for projects.



Different Types of Wi-Fi Modules



ESP-12 Wi-Fi Module

- ESP8266 is a cost-effective, easy-to-operate, compact-sized and low-powered Wi-Fi module.
- It supports both TCP/IP and Serial Protocols.
- It's normally used in IoT-based embedded projects and is considered the most widely used Wi-Fi module because of its low cost and small size.



NodeMCU ESP8266: Version



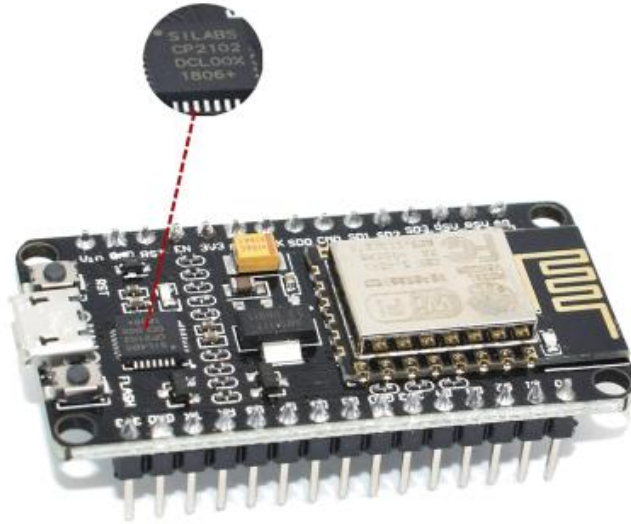
ESP8266 CP2102



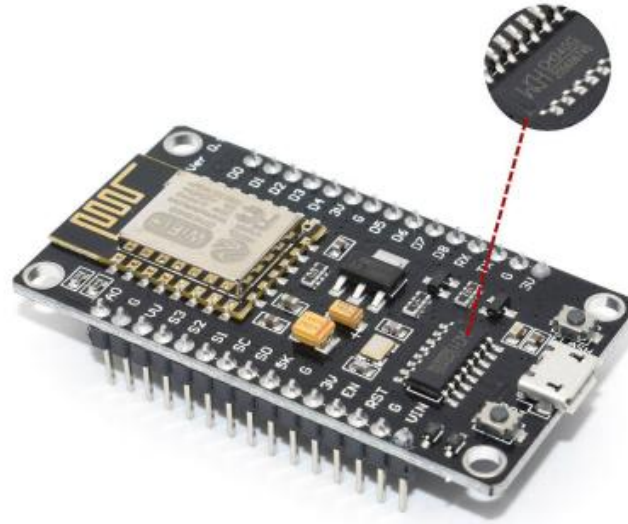
ESP8266 CH340 V3



NodeMCU ESP8266: Versions



ESP8266 CP2102



ESP8266 CH340 V3



NodeMCU ESP8266 V3

- NodeMCU is a low-cost open-source IoT platform based on the ESP8266 Wi-Fi system on a chip.
- NodeMCU Version 3 runs on the ESP-12E (ESP8266MOD) module, and it is easy-to-use development board equipped with analog and digital pins, a USB-to-serial adapter based on CH340G module, and a micro-USB.



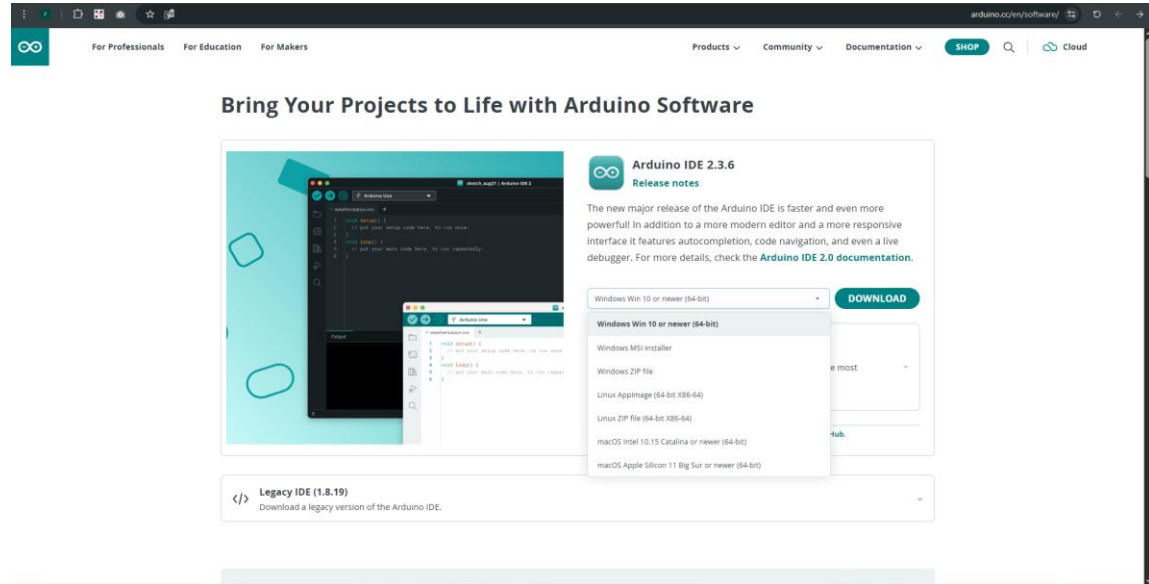
Arduino IDE Download

Step1: Go to this Link <https://www.arduino.cc/en/software/>

Choose your OS:

- Windows
- macOS
- Linux

Download the latest version

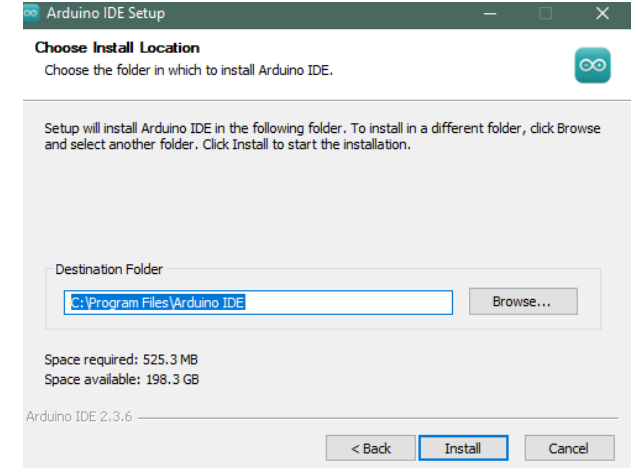
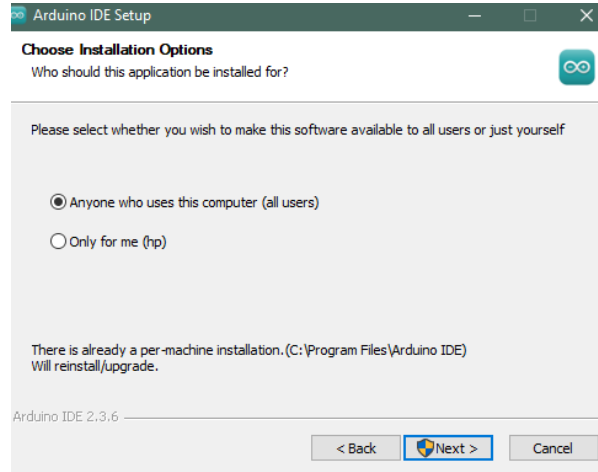


Arduino IDE

❖ How to install Arduino IDE?

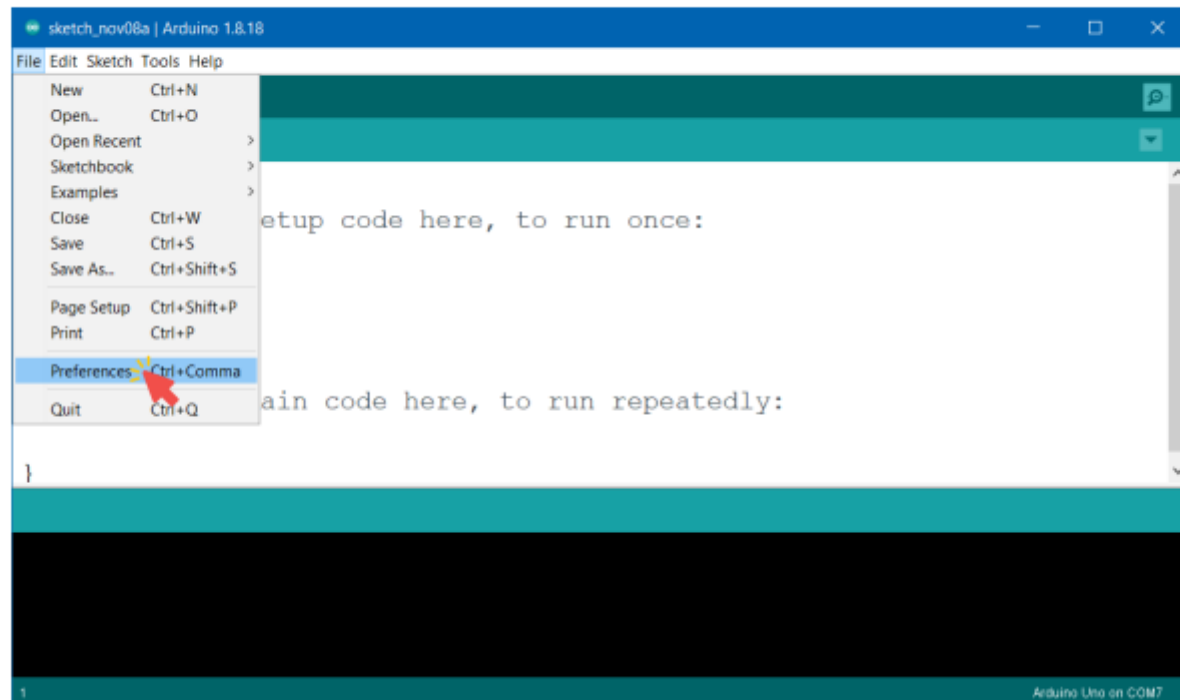
Step2:

- Windows: Run .exe → Next → Install
- macOS: Drag Arduino app → Applications
- Linux: Extract .tar.xz → run install.sh



Arduino IDE

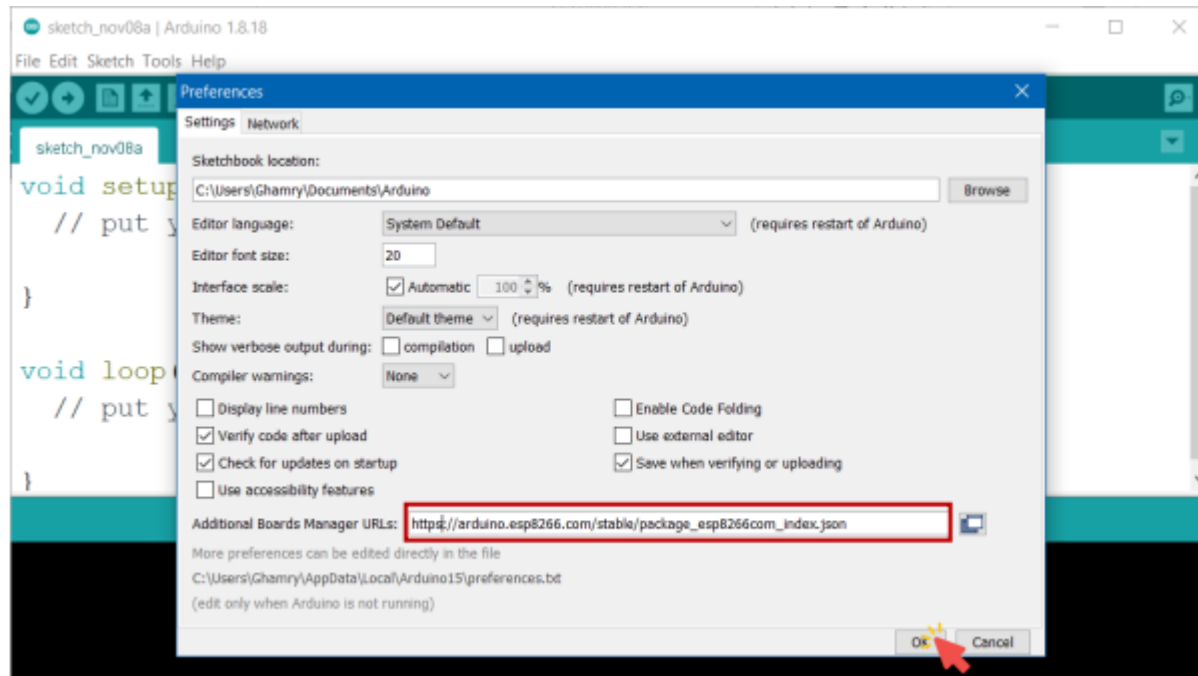
- First, open the **Arduino IDE** and go to **File** → **Preferences**.



Arduino IDE

On the Additional Boards Manager URLs, paste the following URL.

https://raw.githubusercontent.com/espressif/arduino-esp32/ghpages/package_esp32_index.json



Esp32

